

Experiment No: 10

Demonstration of antivirus software and exposure to Malicious Software

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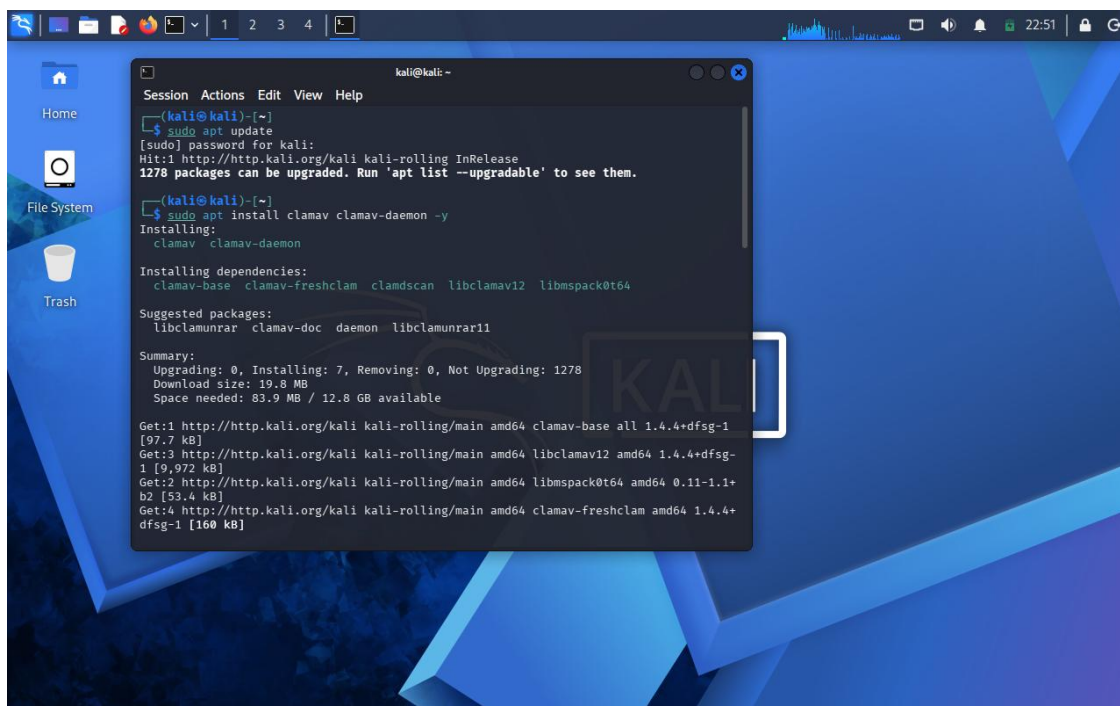
Roll No: A44

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PRN: 2324000718

Step 1: Update System

Step 2 :Install ClamAV

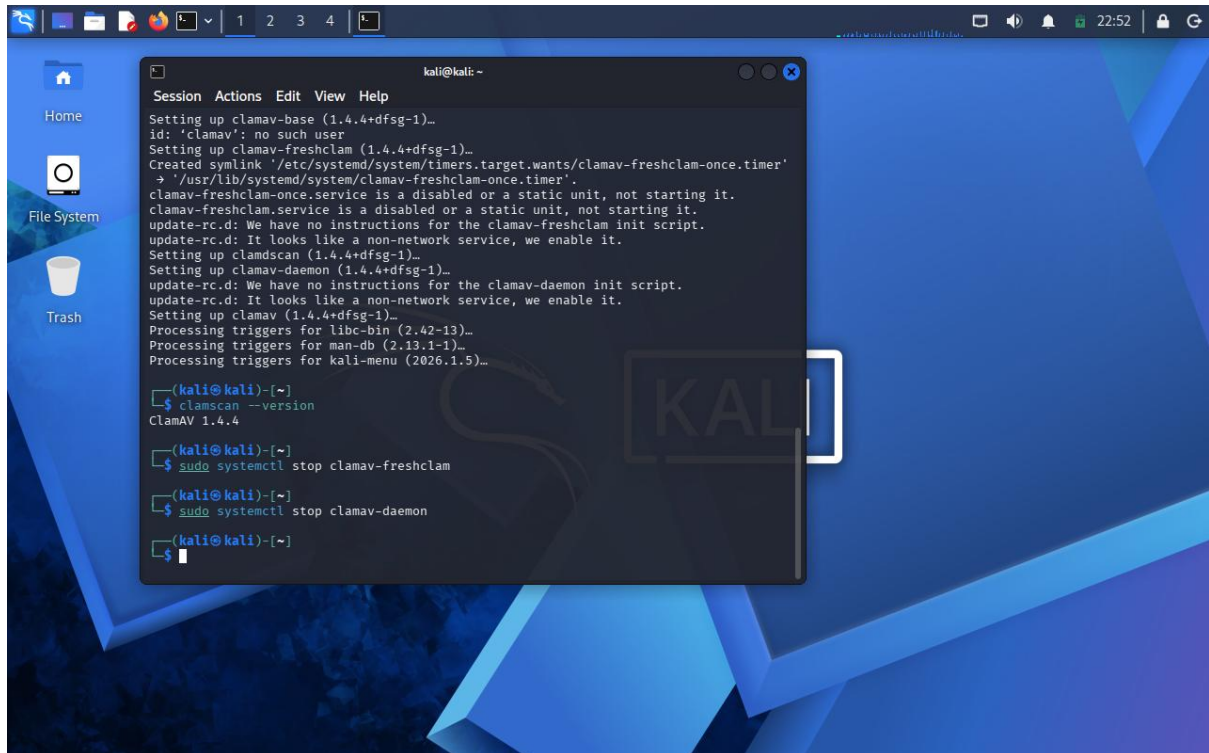


The screenshot shows a Kali Linux desktop environment with a terminal window open. The terminal displays the following commands and output:

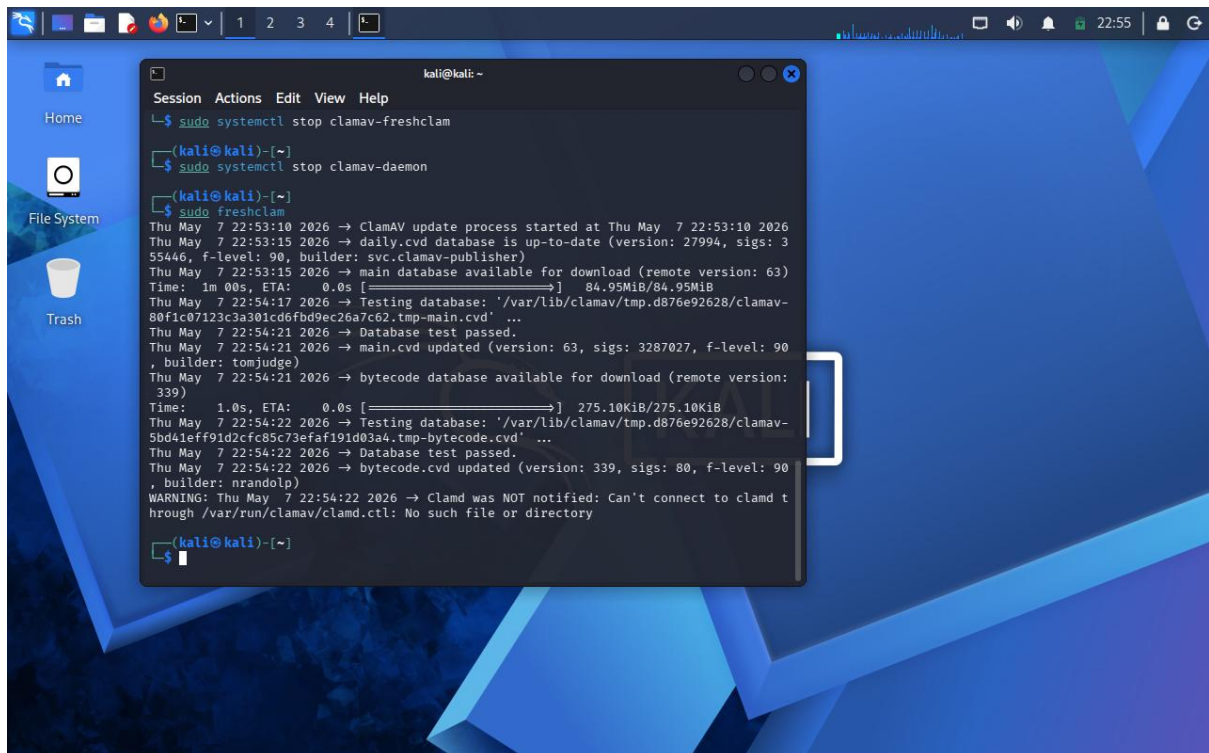
```
kali@kali: ~  
Session Actions Edit View Help  
~(kali@kali)-[~]  
└─$ sudo apt update  
[sudo] password for kali:  
Hit:1 http://http.kali.org/kali kali-rolling InRelease  
1278 packages can be upgraded. Run 'apt list --upgradable' to see them.  
~(kali@kali)-[~]  
└─$ sudo apt install clamav clamav-daemon -y  
Installing:  
clamav clamav-daemon  
  
Installing dependencies:  
clamav-base clamav-freshclam clamdscan libclamav12 libmspack0t64  
  
Suggested packages:  
libclamunrar clamav-doc daemon libclamunrar11  
  
Summary:  
Upgrading: 0, Installing: 7, Removing: 0, Not Upgrading: 1278  
Download size: 19.8 MB  
Space needed: 83.9 MB / 12.8 GB available  
  
Get:1 http://http.kali.org/kali kali-rolling/main amd64 clamav-base all 1.4.4+dfsg-1 [97.7 kB]  
Get:3 http://http.kali.org/kali kali-rolling/main amd64 libclamav12 amd64 1.4.4+dfsg-1 [9,972 kB]  
Get:2 http://http.kali.org/kali kali-rolling/main amd64 libmspack0t64 amd64 0.11-1.1+b2 [53.4 kB]  
Get:4 http://http.kali.org/kali kali-rolling/main amd64 clamav-freshclam amd64 1.4.4+dfsg-1 [160 kB]
```

Step 3 : Verify Installation

- Update Virus Signatures
- Start ClamAV Service



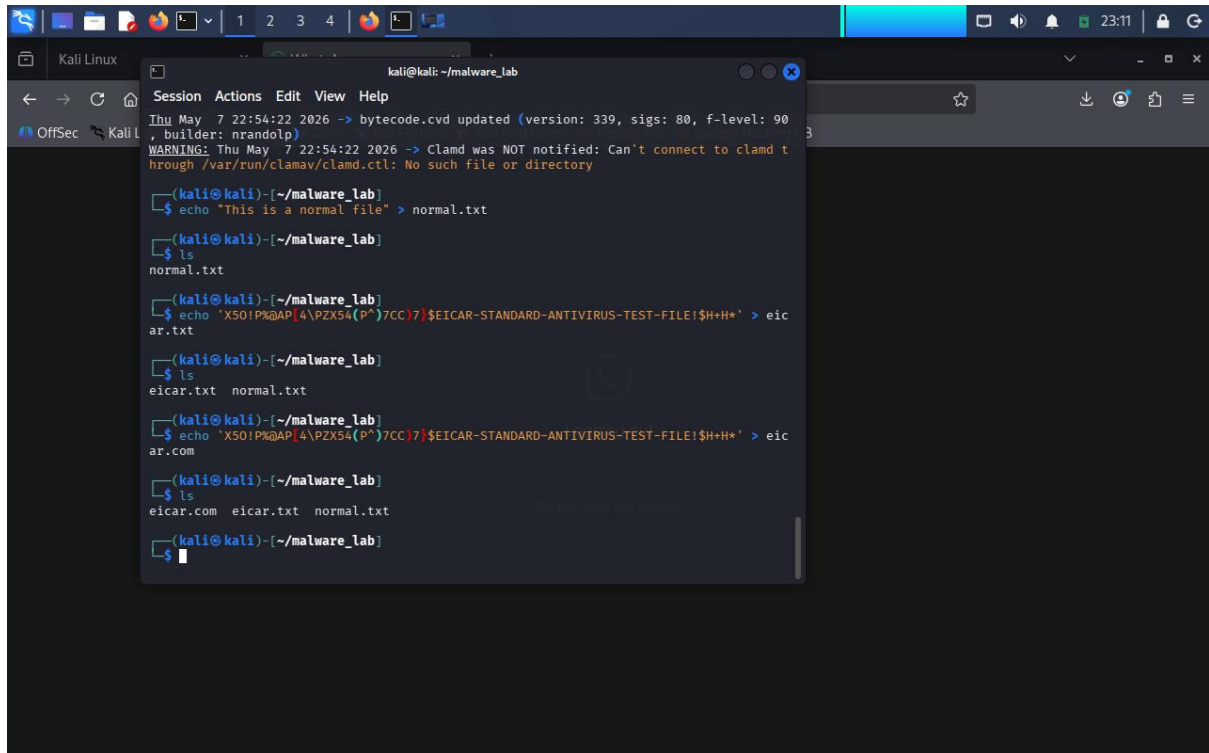
```
kali@kali: ~  
Session Actions Edit View Help  
Setting up clamav-base (1.4.4+dfsg-1)...  
id: 'clamav': no such user  
Setting up clamav-freshclam (1.4.4+dfsg-1)...  
Created symlink '/etc/systemd/system/timers.target.wants/clamav-freshclam-once.timer'  
→ '/usr/lib/systemd/system/clamav-freshclam-once.timer'.  
clamav-freshclam-once.service is a disabled or a static unit, not starting it.  
clamav-freshclam.service is a disabled or a static unit, not starting it.  
update-rc.d: We have no instructions for the clamav-freshclam init script.  
update-rc.d: It looks like a non-network service, we enable it.  
Setting up clamdscan (1.4.4+dfsg-1)...  
Setting up clamav-daemon (1.4.4+dfsg-1)...  
update-rc.d: We have no instructions for the clamav-daemon init script.  
update-rc.d: It looks like a non-network service, we enable it.  
Setting up clamav (1.4.4+dfsg-1)...  
Processing triggers for libc-bin (2.42-13)...  
Processing triggers for man-db (2.13.1-1)...  
Processing triggers for kali-menu (2026.1.5)...  
  
(kali@kali)~  
$ clamscan --version  
ClamAV 1.4.4  
  
(kali@kali)~  
$ sudo systemctl stop clamav-freshclam  
  
(kali@kali)~  
$ sudo systemctl stop clamav-daemon  
  
(kali@kali)~  
$
```



```
kali@kali: ~  
Session Actions Edit View Help  
$ sudo systemctl stop clamav-freshclam  
  
(kali@kali)~  
$ sudo systemctl stop clamav-daemon  
  
(kali@kali)~  
$ sudo freshclam  
Thu May 7 22:53:10 2026 → ClamAV update process started at Thu May 7 22:53:10 2026  
Thu May 7 22:53:15 2026 → daily.cvd database is up-to-date (version: 27994, sigs: 3  
55446, f-level: 90, builder: svc.clamav-publisher)  
Thu May 7 22:53:15 2026 → main database available for download (remote version: 63)  
Time: 1m 00s, ETA: 0.0s [=====] 84.95MiB/84.95MiB  
Thu May 7 22:54:17 2026 → Testing database: '/var/lib/clamav/tmp.d876e92628/clamav-  
80f1c07123c3a301cd6fbd9ec26a7c62.tmp-main.cvd' ...  
Thu May 7 22:54:21 2026 → Database test passed.  
Thu May 7 22:54:21 2026 → main.cvd updated (version: 63, sigs: 3287027, f-level: 90  
, builder: tomjudge)  
Thu May 7 22:54:21 2026 → bytecode database available for download (remote version:  
339)  
Time: 1.0s, ETA: 0.0s [=====] 275.10KiB/275.10KiB  
Thu May 7 22:54:22 2026 → Testing database: '/var/lib/clamav/tmp.d876e92628/clamav-  
5bd41eff91d2cfc85c73efaf191d03a4.tmp-bytecode.cvd' ...  
Thu May 7 22:54:22 2026 → Database test passed.  
Thu May 7 22:54:22 2026 → bytecode.cvd updated (version: 339, sigs: 80, f-level: 90  
, builder: nrandolp)  
WARNING: Thu May 7 22:54:22 2026 → Clamd was NOT notified: Can't connect to clamd t  
hrough /var/run/clamav/clamd.ctl: No such file or directory  
  
(kali@kali)~  
$
```

Step 4 : Create Lab Directory

- Create Normal File
- Create Test Malware



A terminal window titled 'kali@kali: ~/malware_lab' shows the following commands and output:

```
Thu May 7 22:54:22 2026 -> bytecode.cvd updated (version: 339, sigs: 80, f-level: 90, builder: nrandolp)
WARNING: Thu May 7 22:54:22 2026 -> Clamd was NOT notified: Can't connect to clamd through /var/run/clamav/clamd.ctl: No such file or directory

(kali@kali)-[~/malware_lab]
└─$ echo "This is a normal file" > normal.txt

(kali@kali)-[~/malware_lab]
└─$ ls
normal.txt

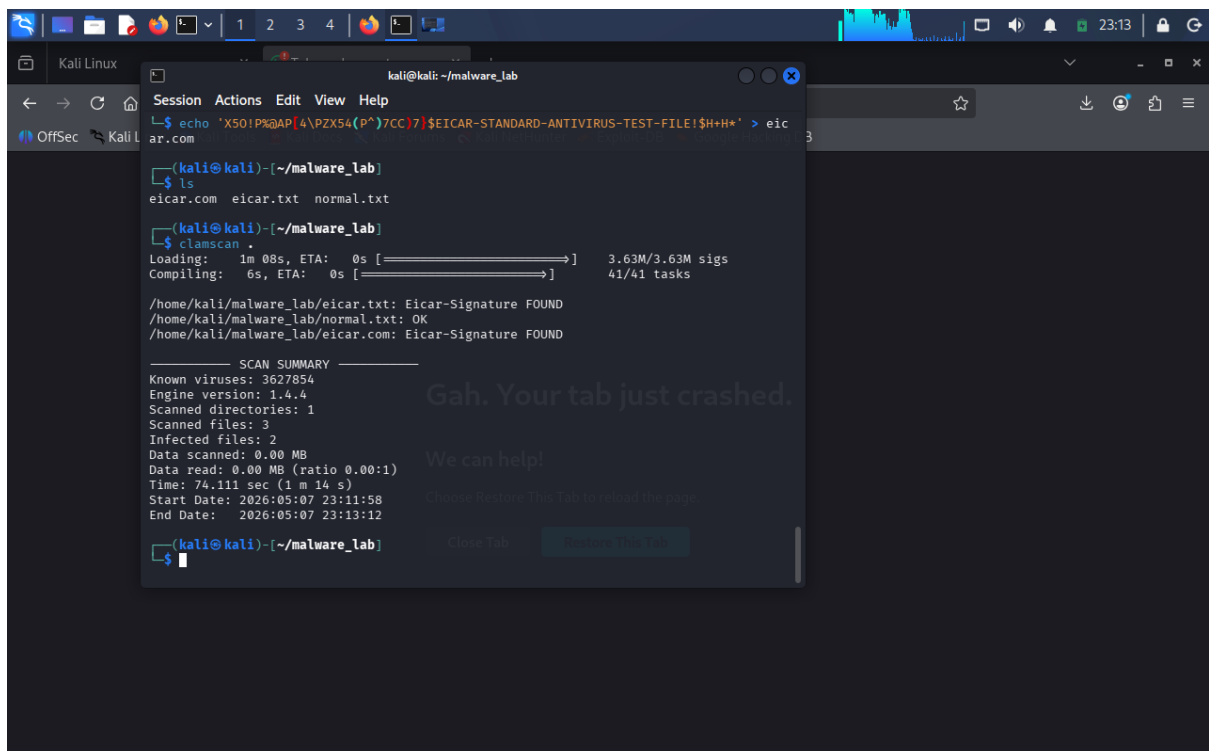
(kali@kali)-[~/malware_lab]
└─$ echo 'X50!P%QAP[4\pZX54(P*)7CC)7)$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*' > eicar.txt

(kali@kali)-[~/malware_lab]
└─$ ls
eicar.txt  normal.txt

(kali@kali)-[~/malware_lab]
└─$ echo 'X50!P%QAP[4\pZX54(P*)7CC)7)$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*' > eicar.com

(kali@kali)-[~/malware_lab]
└─$ ls
eicar.com  eicar.txt  normal.txt
```

Step 5: Scan Files



A terminal window titled 'kali@kali: ~/malware_lab' shows the following commands and output:

```
(kali@kali)-[~/malware_lab]
└─$ echo 'X50!P%QAP[4\pZX54(P*)7CC)7)$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*' > eicar.com

(kali@kali)-[~/malware_lab]
└─$ ls
eicar.com  eicar.txt  normal.txt

(kali@kali)-[~/malware_lab]
└─$ clamscan .
Loading: 1m 08s, ETA: 0s [=====>] 3.63M/3.63M sigs
Compiling: 6s, ETA: 0s [=====>] 41/41 tasks

/home/kali/malware_lab/eicar.txt: Eicar-Signature FOUND
/home/kali/malware_lab/normal.txt: OK
/home/kali/malware_lab/eicar.com: Eicar-Signature FOUND

----- SCAN SUMMARY -----
Known viruses: 3627854
Engine version: 1.4.4
Scanned directories: 1
Scanned files: 3
Infected files: 2
Data scanned: 0.00 MB
Data read: 0.00 MB (ratio 0.00:1)
Time: 74.111 sec (1 m 14 s)
Start Date: 2026:05:07 23:11:58
End Date: 2026:05:07 23:13:12

(kali@kali)-[~/malware_lab]
└─$
```

Overlaid on the terminal is a browser crash message: "Gah. Your tab just crashed. We can help! Choose Restore This Tab to reload the page." with buttons for "Close Tab" and "Restore This Tab".